

WEEK 7 PROJECT SUMMARY

Internship Track: Data Analytics

Project: HealthConnect Clinic Experience Lab

Stage: Week 7 (Testing, Refinement & End-to-End Validation)

1. What I Planned to Test

- Verification of data partitioning and cancellation isolation.
- Arithmetic accuracy and precision of core attendance KPIs.
- Behavioral consistency across boundary conditions and edge cases.
- Dashboard filter integrity and chart visual usability.
- Cross-track alignment between empirical risk brackets and Data Science model predictions.

2. What I Actually Tested

- Executed unit test assertions on 4,737 active and 263 cancelled records.
- Recalculated the baseline no-show rate and reminder attendance lift formulas.
- Tested edge cases: zero-day bookings and patients with 4 or more past no-shows.
- Evaluated chart readability and stakeholder usability.
- Cross-tested empirical lead-time brackets against Data Science model risk scores.

3. The Most Important Testing Results

- Dataset Partitioning Passed: 4,737 active records + 263 cancelled records reconcile perfectly to 5,000 rows.
- KPI Accuracy Verified: Overall no-show rate is exactly 51.15% (2,423 / 4,737) and reminder lift is exactly +4.78%.
- Zero Lead Time Validation: Same-day bookings (81 rows) had a failure rate of only 25.93%.
- Chronic Failure Validation: Patients with 4+ prior no-shows (15 rows) had a failure rate of 73.33%.

4. Issues or Weaknesses Identified

- Initial Week 6 charts showed bar values but lacked clinic average benchmark lines.
- Without reference lines, stakeholders could not quickly identify which booking windows breached acceptable limits.

5. Improvements and Refinements Made

- Added a blue dashed benchmark line at 51.15% (clinic average) on the lead-time chart.
- Added a red dashed benchmark line at 54.17% on the follow-up channel chart.

- Re-mapped color palettes to emphasize high-risk zones (>55%) and safe zones (<35%).
- Added direct numeric percentage badges to all chart bars.

6. Retesting Results

- Retested visual charts with updated annotations.
- Stakeholders can now immediately spot that lead-time risk breaches the clinic average after 30 days.
- All 7 test scenarios achieved a 100% final pass status.

7. Which Track(s) You Collaborated With

Data Science Track HC POD 4 (Collaborator: M'bengue Mama).

8. What Was Tested Collaboratively

- Validated segment performance from the Data Science held-out test set (n=1,903) against Data Analytics population benchmarks across lead times, reminder channels and patient history.
- Analyzed model error patterns (570 False Positives vs. 188 False Negatives) and the impact of the 0.40 decision threshold.

9. What Changed as a Result

- Confirmed near-perfect segment concordance (e.g., 31+ day failure rate is 63.7% in the test set vs. 64.0% in my analytics).
- Established that high recall (80.54% overall, 99.8% on 31+ days) justifies the false positive rate because the model is restricted to friendly outreach.
- Formalized the clinic decision rule: model risk scores will guide reminder prioritization and will never be used to penalize patients.

10. Key Findings and Validation Outcomes

- Test Set Alignment: Data Science test metrics mirror my population numbers within 0.2% to 1.7%.
- Catching High-Risk Visits: The refined model catches 99.8% of no-shows in the critical 31+ day booking window.
- Clinical Safety Confirmed: Error analysis validated that outreach false positives cause no harm to attending patients.

11. Major Challenges Encountered

- Writing automated assertion tests that account for floating-point precision differences.
- Ensuring cancelled visits did not accidentally re-enter filtered attendance queries.

12. Important Decisions Made

- Benchmarking Decision: Made baseline reference lines mandatory on all future clinic dashboards.

- Tiered Scheduling Policy: Confirmed that bookings over 21 days must receive two-stage automated SMS reminders.
- Standby Reallocation: Recommended releasing unconfirmed long-lead slots to walk-in patients 24 hours prior.

13. Remaining Limitations

- The dataset does not track real-time SMS delivery confirmations or message open status.
- All data remains adult-only (18+) and does not reflect pediatric clinic operations.
- External factors such as weather, public transit disruptions and income remain unrecorded.

14. Remaining Issues or Dependencies

- Awaiting final end-to-end pipeline deployment from the Machine Learning Engineering track.
- Need final confirmation from Project Management on presentation slide allocation.

15. Contribution to the Overall HealthConnect Project

- Delivered 100% verified, zero-error KPI metrics for executive decision-making.
- Validated that empirical analytics and machine learning model predictions agree.
- Provided refined, stakeholder-ready visualizations that clearly highlight operational risk.

16. What Must Be Completed Before Week 8

- Assemble the final integrated executive presentation slide deck.
- Consolidate all weekly documentation and notebooks into a portfolio-ready repository.
- Finalize executive speaking points for the final project presentation.